■ 2.6.3 Advanced Topic: Outputting Internal Representations

The form in which *Mathematica* prints an expression often bears little relation to the form in which the expression is actually stored. For most purposes, you will never need to know exactly how expressions are stored. However, if you want to manipulate the *structure* of an expression, particularly as part of a program, you may need to look at its internal form.

`FullForm[`*expr*`]`	the explicit form of *expr* in functional notation
`TreeForm[`*expr*`]`	a representation of the expression tree for *expr*
`PrintForm[`*expr*`]`	the internal representation of the print form for *expr*

Output forms that reveal internal representation.

Here is an expression in standard *Mathematica* output form.

```
In[1]:= x^2/3 + 1/y^2

         2
        x     -2
Out[1]= -- + y
        3
```

Here is the same expression in an explicit functional notation that mirrors the internal representation of the expression.

```
In[2]:= FullForm[%]
Out[2]//FullForm=
  Plus[Times[Rational[1, 3], Power[x, 2]], Power[y, -2]]
```

Here is a representation of the expression as a tree structure.

```
In[3]:= TreeForm[%]
Out[3]//TreeForm=
  Plus[|                             , |           ]
       Times[|             , |        ]  Power[y, -2]
             Rational[1, 3]  Power[x, 2]
```

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